Questions by topic: Mineral resources (the lithosphere) - Set 1 (75 marks)

Answer all questions in the spaces provided.

0 1 . 1 Describe how hydrothermal deposition produces exploitable concentrations of metal ores.

[4 marks]

0 1 . 2 Explain how Proterozoic marine sediments produced the world's major iron ore deposits.

[2 marks]

0 1 . 3 Explain how alluvial (physical sediment) processes separate and concentrate minerals such as gold and tin ore.

[2 marks]

0 1 . 4 Give **two** examples of mineral deposits formed by biological sedimentation and state their origin.

[2 marks]

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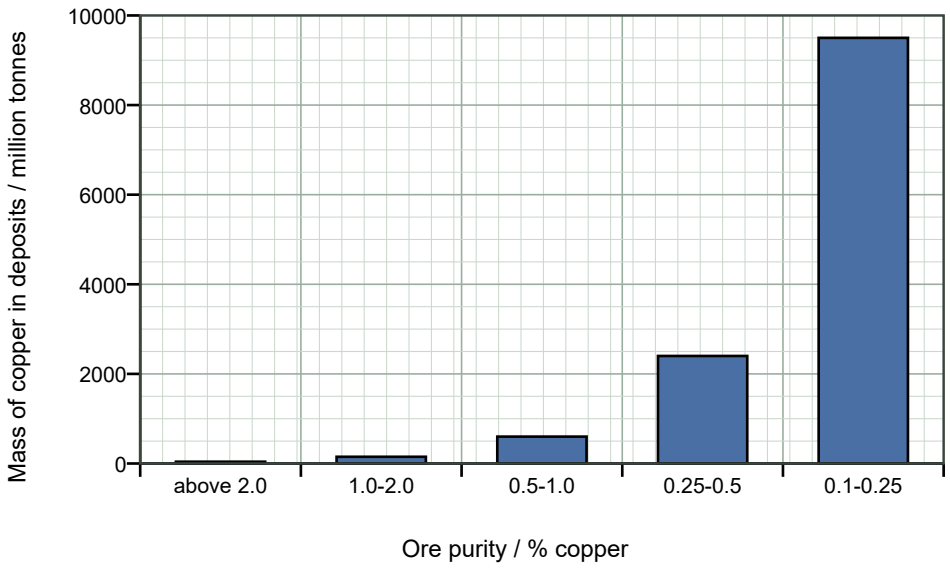
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0 2 Figure 1 shows the mass of copper contained in the Earth's crust at different ore purities, illustrating Lasky's principle.

Figure 1



0 2 . 1 State Lasky's principle.

[1 mark]

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0 2 . 2 Use Figure 1 to calculate how many times more copper is contained in deposits of 0.25-0.5% purity than in deposits above 2.0%.

[1 mark]

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Answer [ ] times

0 2 . 3 Explain the difference between the **reserves** and the **resource** of a mineral.

[3 marks]

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0 2 . 4 Explain, using Lasky's principle, why the ability to exploit slightly lower grade ores greatly increases reserves.

[2 marks]

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0 2 . 5 Explain why the main problem for future mineral supplies is not the quantity that exists in the crust.

[3 marks]

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0 3 Table 1 shows some techniques used to locate mineral deposits.

0 3 . 1 Complete Table 1 by adding the name of the technique or a description of how it works.

[4 marks]

Table 1

Technique	How it works
Gravimetry	
	Detects rocks that are more magnetic, such as magnetite iron ore or ores of tungsten and cobalt
Seismic survey	
	Measures how difficult it is for electricity to pass through rocks; sedimentary rocks with high water content have lower values than igneous rocks

0 3 . 2 Explain why trial drilling is usually carried out only after remote sensing surveys.

[3 marks]

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0 3 . 3 Explain how infrared spectroscopy from satellites can help identify mineral deposits.

[2 marks]

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0 3 . 4 Give **one** advantage of portable field analysis equipment such as X-ray fluorescence.

[1 mark]

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0 4 Table 1 shows information about two copper deposits, X and Y, being assessed for mining. The current cut-off ore grade for copper is 0.4%.

Table 1

Feature	Deposit X	Deposit Y
Ore grade / % copper	0.9	0.35
Depth of deposit / m	350	40
Overburden	Hard igneous rock	Loose glacial sediment
Distance to nearest railway / km	120	8
Annual rainfall / mm	2 100	500
Chemical form	Copper sulfide	Copper sulfide

0 4 . 1 Explain what is meant by the **cut-off ore grade** and state which deposit is currently included in the copper reserves.

[2 marks]

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0 4 . 2 Use Table 1 to discuss the factors that affect the viability of mining each deposit.

[6 marks]

0 4 . 3 State **two** changes that would decrease the cut-off ore grade.

[2 marks]

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0 5 . 1 Explain why mineral extraction causes more land-use conflict than most other human activities.

[2 marks]

0 5 . 2 Describe how each of the following environmental impacts of an operating mine can be reduced.

[5 marks]

Dust

Noise

Turbid drainage water

Acidic leachate from spoil

Loss of amenity

0 5 . 3 Explain how spoil heaps can be made stable and describe what happened at Aberfan in 1966.

[3 marks]

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0 6 A sand and gravel pit near a town is due to close. Three options for the site are proposed: a wetland nature reserve, housing, or return to farmland.

0 6 . 1 Explain why flooded sand and gravel pits are often developed as wetland nature reserves.

[2 marks]

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0 6 . 2 Explain the conditions that would need to be met for the site to be used for housing or farmland.

[2 marks]

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0 6 . 3 Explain why moving the wildlife from a site before mining begins is rarely completely successful.

[1 mark]

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0 7 . 1 Describe how bioleaching is used to extract copper from low-grade ores or spoil heaps, and explain why it may be more sustainable than smelting.

[5 marks]

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0 7 . 2 Describe the process of phytomining.

[3 marks]

0 7 . 3 Explain how polymer adsorption may provide a low-energy source of uranium in the future.

[2 marks]

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0 8 . 1 Describe polymetallic (manganese) nodules and explain why they are not yet exploited on a large scale.

[3 marks]

0 8 . 2 Explain the environmental impacts of mining polymetallic nodules.

[3 marks]

0 8 . 3 Explain why rare earth metals are a supply concern and describe one new extraction technique.

[4 marks]

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END OF QUESTIONS

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